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(57) Abstract :

The design and performance analysis of a compact dual-band coplanar waveguide (CPW) fed planar antenna for X and Ku band for wireless communication in the domain of satellite applications. The proposed antenna is designed and simulated using the High Frequency Structure Simulator (HFSS) to achieve dual-band operation at 7.6 GHz and 15.5 GHz. The antenna is fabricated on an RT Duroid substrate with a relative permittivity (ϵ_r) of 2.2, which provides low dielectric loss and enhanced radiation efficiency. The antenna structure employs a planar radiating element integrated with a CPW feed line to achieve improved impedance matching and stable radiation characteristics. The simulated results show good antenna performance with return loss values of -20.02 dB at 7.6 GHz and -22.36 dB at 15.5 GHz, while the corresponding VSWR values are 1.22 and 1.17, indicating efficient power transfer at the operating frequencies. The impedance bandwidths (S11 -10 dB) are observed to be 940 MHz (7150–8090 MHz) and 1220 MHz (15070–16290 MHz) around the respective resonant frequencies. In addition, the antenna achieves a peak gain of 5 dB with stable radiation patterns. The obtained results demonstrate that the proposed antenna is suitable for X-band and Ku-band for wireless communication, satellite and radar applications.

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